Summary in Graph

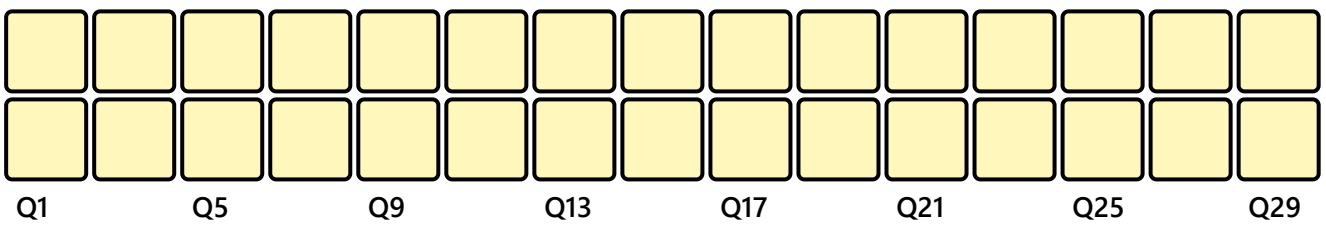
Exam Summary (GATE Overflow CS Test Series 2026 | Calculus | Shallow Test 1)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (2) out of 18 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

If $p(x) = x^{200} - x^{199} + x^{198} - x^{197} + \cdots - x + 1$, then $p'(1)$ is equal to:

A. 201
B. 199
C. 100
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider the function $f(x) = \sin x$ on the interval $[0, \pi]$ for which value of c does $Lagrange's$ Mean Value Theorem apply to $f(x)$? A. $\frac{\pi}{4}$ B. $\frac{\pi}{2}$ C. $\frac{2}{\pi}$ D. $\pi - 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2 + \sin x$. Which of the following statements about the behavior of $f(x)$ is correct?

A. $f(x)$ is a decreasing function for all $x \in \mathbb{R}$.
B. $f(x)$ is not monotonic for small values of $x > 0$.
C. $f(x)$ attains a local maximum at $x = 0$.
D. $f(x)$ does not attain a local minimum at $x = 0$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

For the following function $f(x) = \sec x \left(4 \sin 2x - 6 \cos^2 x\right)$ determine its least value.

A. -10

B. 10

C. $2\sqrt{5}$

D. $-5\sqrt{2}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

For which positive integer n does the following equation hold?

$$\lim_{x \to 1} \frac{\ln(x^n + 1) - \ln(2)}{x - 1} = 3$$

A. 2
B. 4
C. 6
D. 8

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

Determine the value of the following expression:

$$\lim_{x \to 0} \frac{x^2 + 2x - 2}{x^3 + x}$$

Which of the following options is correct?

A. 1
B. 0
C. $\frac{1}{2}$
D. The limit does not exist

Your Answer: Correct Answer: D Not Attempted Time taken: 00min 00sec Discuss

Q #7 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

Evaluate the following limit involving the difference of tangents:

$$\lim_{x \to 0} \frac{\tan(4 + x) - \tan(4 - x)}{x}$$

What is the value of this limit?

A. $2 \sec^2 4$
B. $4 \sec^2 4$
C. $4 \sin 4$
D. Does not exist

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #8 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

Evaluate the limit as $x \to \frac{\pi}{2}$ for the expression:

$$\lim_{x \to \frac{\pi}{2}} \frac{\sqrt{1 - \cos(4x)}}{\sqrt{2} \cdot (\frac{\pi}{2} - x)}$$

A. 1
B. 2
C. -2
D. Limit does not exist.

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #9 Multiple Choice Type Award: 1 Penalty: 0.33 Calculus

Determine the value of the limit:

$$\lim_{x \to 1} \frac{1 - \frac{1}{x}}{\sin 2\pi(x - 1)}$$

A. 2π
B. 1
C. $\frac{1}{2\pi}$
D. -1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Determine the value of the limit as x approaches 4 for the expression:

$$\lim_{x \to 4} \frac{x-4}{|4-x|}$$

A. 1
B. Limit does not exist
C. -1
D. 0

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Determine the value of the following limit:

$$\lim_{x \to \infty} a^x \sin\left(\frac{2b}{a^x}\right), \quad \text{where } a, b > 1.$$

A. $\log a$
B. b
C. $2b$
D. $a \log_e b$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $g(x)$ be the function defined as

$$g(x) = \begin{cases} \frac{m \sin x}{x-\pi}, & x \neq \pi, \\ 2, & x = \pi. \end{cases}$$

If $\lim_{x \to \pi} g(x) = g(\pi)$, the value of m is

A. 2
B. $2\sqrt{2}$
C. $\sqrt{2}$
D. -2

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Find k such that, $\lim_{x\to 2} f(x)$ exists, where $f(x) = \begin{cases} 4x+3, & \text{if } x \le 2 \\ x+k, & \text{if } x > 2 \end{cases}$ A. 2 B. -2 C. 5 D. 9

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Evaluate the following limit, where $\lfloor x \rfloor$ represents the floor function:

$$\lim_{x\to 2} \frac{\log(\lfloor x-1 \rfloor)}{\lfloor x-2 \rfloor}$$

A. 1
B. 0
C. -1
D. Limit does not Exist.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Given that $f(x) = \dfrac{4-8x}{3x+4}$, let:

$$l = \lim_{x\to 2} f(x) \quad \text{and} \quad m = \lim_{x\to 0} f(x)$$

Find the equation whose roots are $\dfrac{1}{l}$ and $\dfrac{1}{m}$.

A. $6x^2 - x - 5 = 0$
B. $6x^2 + x - 5 = 0$
C. $6x^2 - x + 5 = 0$
D. $6x^2 + x + 5 = 0$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Numerical Type | Award: 2 | Penalty: 0 | Calculus

If

$$\lim_{x\to 0^+} kx \cdot \frac{1}{\sqrt{\sin^2 x}} = \lim_{x\to 0^+} x \cdot \operatorname{cosec}(kx),$$

what is the absolute value of k ?

Your Answer: | Correct Answer: 1:1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

If $f(x) = \dfrac{1}{3x^2 + 5x + 2}$, then its maximum value is: A. 0.5 B. -12 C. 12 D. It does not have a maximum value

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The minimum value of the function

$$f(x) = \frac{2x^2}{5} + \frac{1}{x} + 6$$

occurs at:

A. $x = 1$
B. $x = \sqrt[3]{\frac{5}{4}}$
C. $x = 2$
D. $x = -1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f(x)$ be the function defined as

$$f(x) = \frac{x}{\log_e x} \quad \{\text{for} \quad x > 0,\ x \neq 1.$$

f has:

A. local maximum
B. local minimum
C. no local maximum
D. no extremum

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 2 | Penalty: 0 | Calculus

The maximum value of the function

$$f(x) = x^3 - 6x^2 + 9x + 4$$

in the interval $0 \leq x \leq 4$ is _____.

Your Answer: | Correct Answer: 8:8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

When trying to identify the maximum of a continuous function $f(x)$, the necessary and sufficient conditions that must hold at a point x_0 for it to be a local maximum are:

A. $f'(x_0) > 0$ and $f''(x_0) = 0$
B. $f'(x_0) < 0$ and $f''(x_0) = 0$
C. $f'(x_0) = 0$ and $f''(x_0) < 0$
D. $f'(x_0) = 0$ and $f''(x_0) > 0$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Calculus

Consider the function

$$f(x) = \frac{x^2 - 4}{x^2 + 4}$$

Find the value of x in the interval $-20 \leq x \leq 20$ where $f(x)$ attains its minimum.

Your Answer: | Correct Answer: 0:0 | Not Attempted | Time taken: 00min 03sec | Discuss

Q #23 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Which of the following functions is strictly increasing for $x \in \mathbb{R}$?

A. $f(x) = x^3 - 3x + 1$
B. $f(x) = e^x$
C. $f(x) = \frac{x^2}{x - 1}$
D. $f(x) = -x^2 + 4x - 5$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

For the function $f(x) = x^2 - 4x + 3$, determine the value of c in the interval $[1, 3]$ that satisfies the conditions of $Rolle's Theorem.$

A. $c = \frac{1}{2}$
B. $c = 1$
C. $c = 2$
D. $c = -1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Compute the value of

$$\int_0^{\frac{\pi}{2}} \frac{\sin^6 x}{\cos^6 x + \sin^6 x} dx$$

A. π
B. $\frac{\pi}{4}$
C. 1

D. $\frac{\pi}{2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Which of the following function(s) is/are not continous at $x = 0$? (Mark all the appropriate choices)

A. $f(x) = \begin{cases} \frac{\sin 3x}{x}; & \text{when } x \neq 0 \\ 3; & \text{when } x = 0 \end{cases}$

B. $f(x) = \begin{cases} e^{\frac{1}{x}}; & \text{when } x \neq 0 \\ 1; & \text{when } x = 0 \end{cases}$

C. $f(x) = \begin{cases} \frac{1-\cos x}{x^2}; & \text{when } x \neq 0 \\ 1; & \text{when } x = 0 \end{cases}$

D. $f(x) = \begin{cases} \frac{x-|x|}{2}; & \text{when } x \neq 0 \\ 2; & \text{when } x = 0 \end{cases}$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Evaluate the integral

$$I = \int_0^1 \frac{\sqrt{\tan^{-1}(x)}}{1+x^2}\, dx$$

A. $\frac{\pi^{3/2}}{12}$ B. $\frac{\pi^2}{8}$ C. $\frac{\pi}{4}$ D. $\frac{\pi}{6}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Evaluation of the following integral

$$I = \int_1^3 \frac{\cos(\log x)}{x}\, dx$$

gives___.

A. $\sin(\log 3)$
B. $\cos(\log 3)$
C. $\sin(\log 2)$
D. $\cos(\log 2)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Numerical Type | Award: 2 | Penalty: 0 | Calculus

The Value of the following integral

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (2\sin|x| + \cos|x|)\ dx$$

is_____.

Your Answer: | Correct Answer: 6:6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Evaluate the integral

$$I = \int_0^1 \frac{1-x}{1+x}\, dx$$

A. $\log(2) - 1$
B. $2\log(2) - 1$
C. $\log(2)$
D. $2\log(2)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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